Environment

Environmental trigger factors include exposure to extremes of temperature (hot and cold air), temperature changes, caffeine, alcohol, hot and spicy foods, sunlight, exercise, acute psychological stresses, menstruation, demodex mites, and certain medications. There is controversy regarding whether *H. pylori* should be considered a trigger factor for rosacea.

Cytokines

Due to an impaired permeability barrier in the stratum corneum, the release of various cytokines such as tumour necrosis factor α (TNF-α), IL-1, and IL-6 is triggered, leading to cutaneous inflammation—possibly as an attempt by the epidermis to initiate self-repair. These pathomechanisms can lead to neurological symptoms such as stinging and burning sensations.

Antimicrobial Peptides

Recently, overproduction of antimicrobial peptides (AMPs) has been identified as a cause of rosacea lesions. AMPs include defensins and cathelicidins, such as LL-37.

Toll-like Receptors

Keratinocytes express Toll-like receptors to sense pathogen-associated molecular patterns (PAMPs). Toll-like receptor 2 (TLR2) and Toll-like receptor 4 (TLR4) have been shown to be overexpressed in rosacea skin.

Adaptive Immune Cells

T cells (especially Th1/Th17-polarized immune cells), macrophages, mast cells, and neutrophils are present in all subtypes I–III of rosacea. Neutrophils are assumed to play an essential role in the pathogenesis of rosacea. Reactive oxygen species (ROS) and other proteases produced by neutrophils damage blood vessels, causing inflammation and angiogenesis, which can lead to telangiectasias. Recently, involvement of B cells in the pathomechanism of rosacea has also been demonstrated.

Mites

Demodex mites, although normally present in adult skin, are found in increased numbers in many rosacea patients. A reduction in Demodex mite density correlates with improvement in rosacea symptoms.

Blood Vessels

Vascular dilation results in erythema, which may appear transiently as facial flushing. In rosacea, the papillary dermal vasculature is dilated and skin perfusion is increased.

Nerves

Sensory nerve endings release vasoactive neuropeptides via transient receptor potential (TRP) channels. Two calcium channel blockers and members of the TRP family—transient receptor potential vanilloid 1 (TRPV1) and transient receptor potential ankyrin 1 (TRPA1)—are of particular interest in rosacea, as they mediate inflammatory responses. These channels are activated by spices, alcohol consumption, and temperature changes. Upon activation, TRPV1 and TRPA1 induce the release of neuropeptides such as pituitary adenylate cyclase-activating polypeptide (PACAP), substance P (SP), and calcitonin gene-related peptide (CGRP), which collectively cause flushing and erythema through vasodilation. PACAP, SP, and CGRP also promote inflammation by activating mast cells, macrophages, neutrophils, and T cells.

Due to the multifactorial pathogenesis of rosacea, which is difficult to target therapeutically, current treatment strategies focus on symptomatic suppression of inflammation and reduction of disfiguring features. Clinically, four subtypes of rosacea are differentiated based on observable features and severity.

Clinical Manifestations and Subtypes of Rosacea

Patients with subtype I (erythematotelangiectatic rosacea, ETR) present with centrofacial erythematous macules due to dilated facial capillaries, particularly on the nose and cheeks. They may experience episodes of transient erythema (flushing) or non-transient (persistent) erythema. ETR flares result from acute vasodilation and innate inflammation. With chronic vascular dilation, persistent facial erythema develops. Additionally, chronic edema may occur due to extravascular fluid leakage and perivascular inflammation.